

Name: Maira Rizwan

Registration Number: FA24-B1SE-074

Section: B

GitHub Username: mairairizwan74

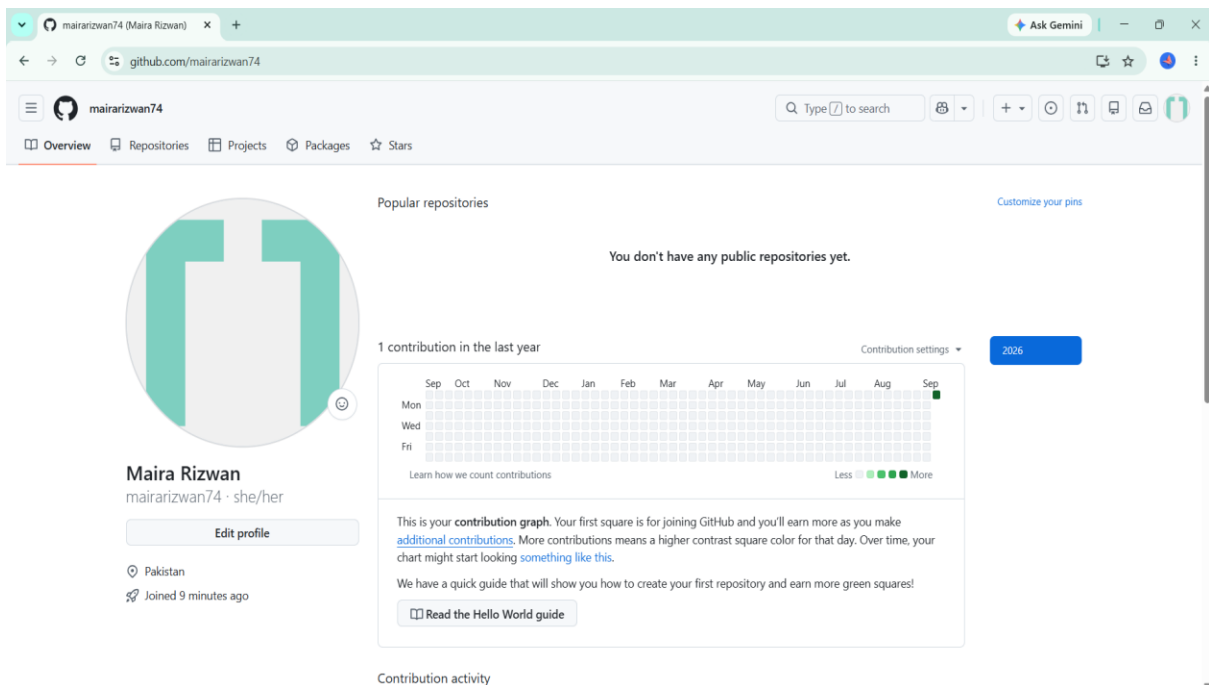
LAB 01

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

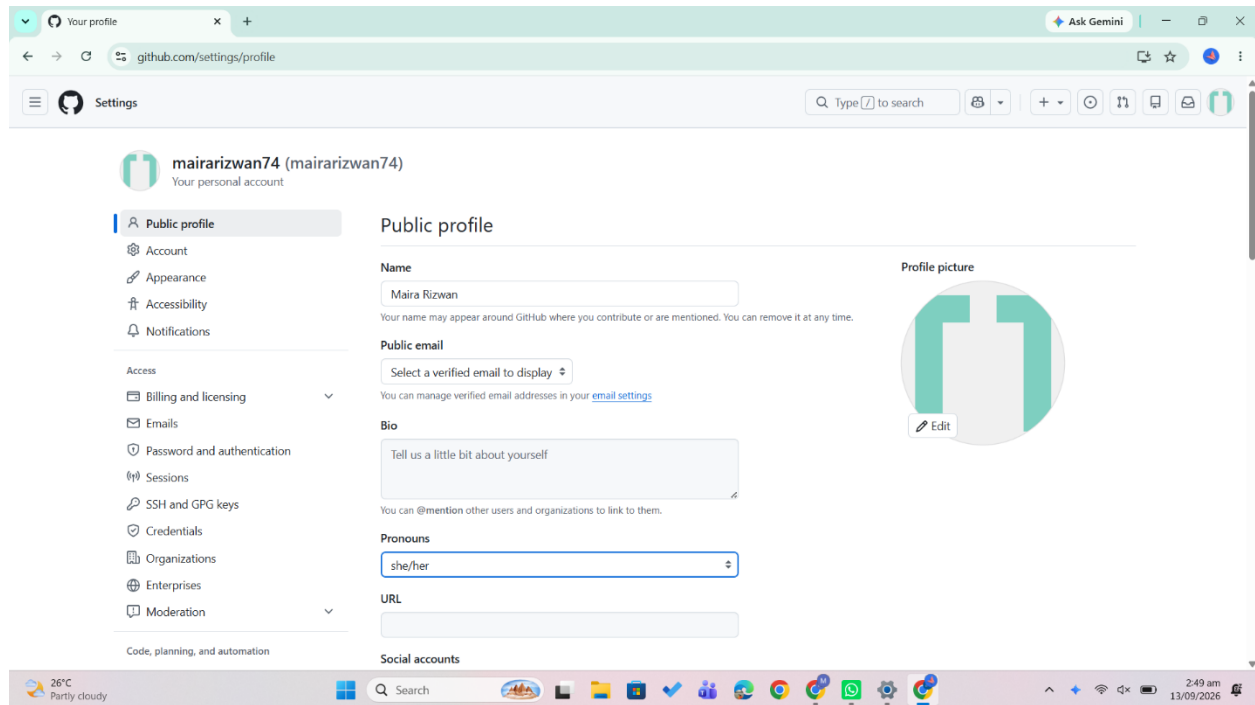
Task 1: Create and Configure Your GitHub Account

Screenshots:

- github_profile.png



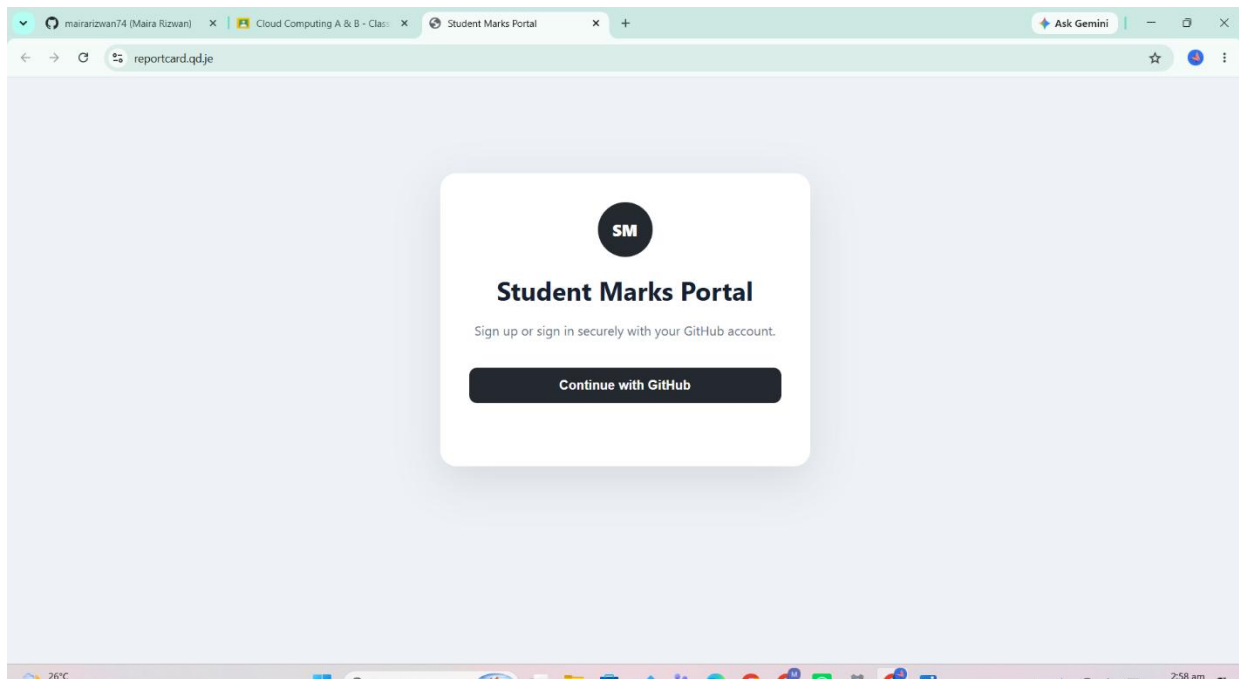
- github_actual_name.png



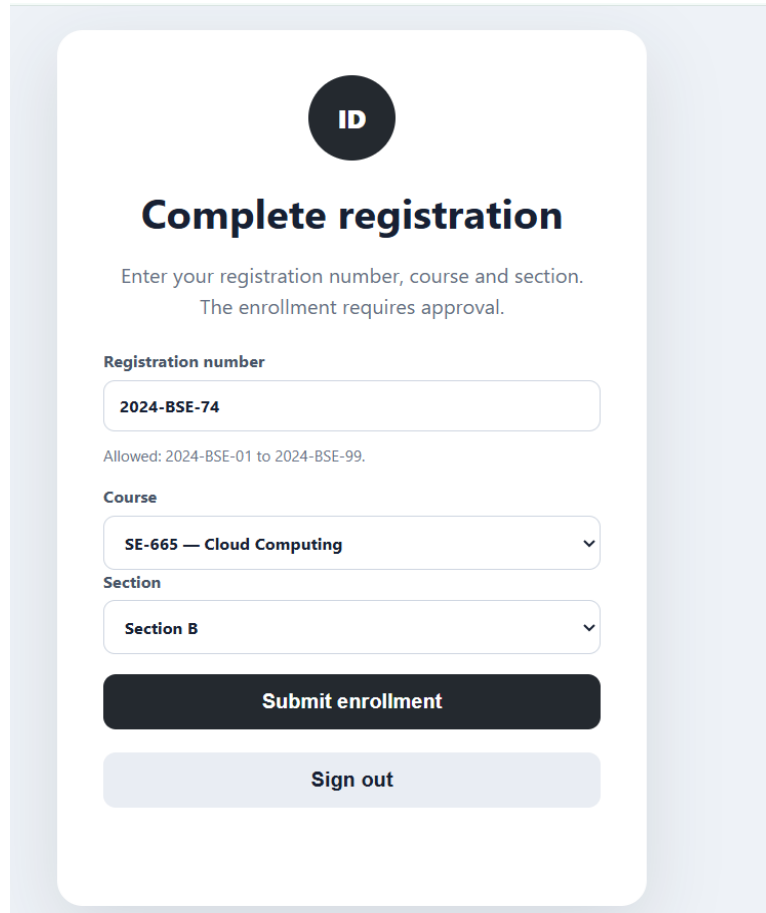
Task 2: Register in the Student Marks Portal

Screenshots:

- portal_login_page.png



- portal_github_profile.png



A registration form titled "Complete registration" with a dark blue header containing an "ID" icon. The form asks for a registration number, course, and section. The registration number is "2024-BSE-74", the course is "SE-665 — Cloud Computing", and the section is "Section B". There are "Submit enrollment" and "Sign out" buttons.

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-74

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

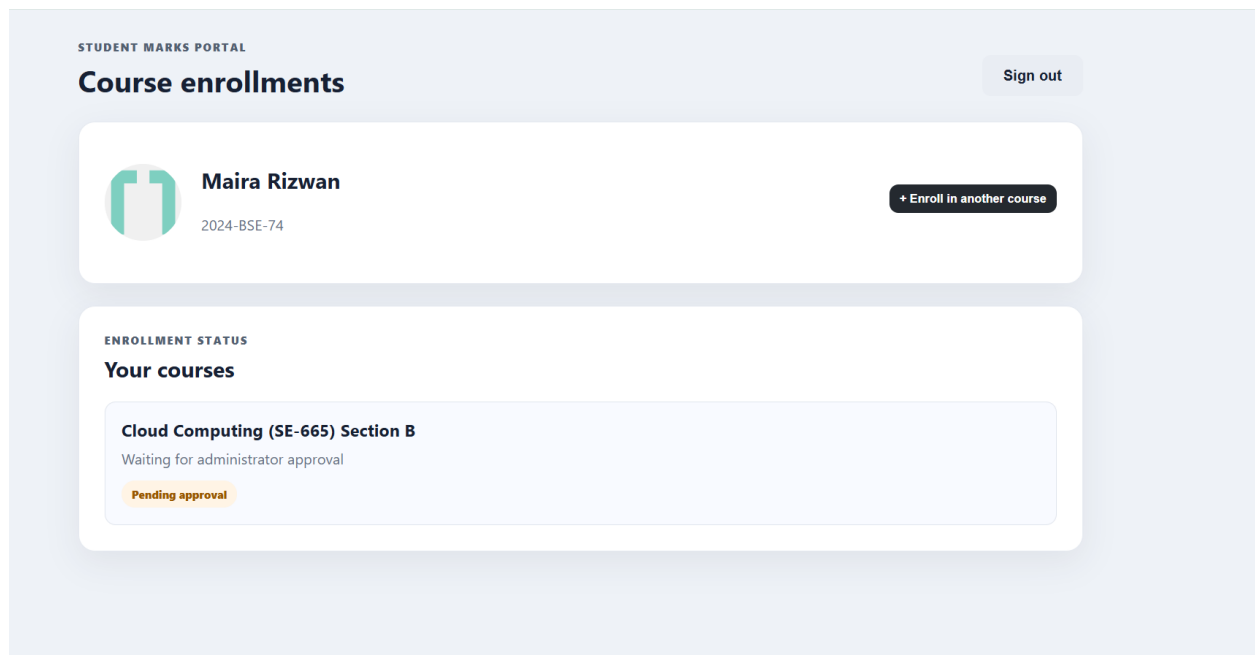
Section

Section B

Submit enrollment

Sign out

- portal_enrollment_submitted.png



A page titled "Course enrollments" with a "Sign out" button. It shows a user profile for Maira Rizwan (2024-BSE-74) and a list of courses. The course "Cloud Computing (SE-665) Section B" is listed with a status of "Waiting for administrator approval" and a "Pending approval" tag.

STUDENT MARKS PORTAL

Course enrollments

Sign out

Maira Rizwan
2024-BSE-74

+ Enroll in another course

ENROLLMENT STATUS

Your courses

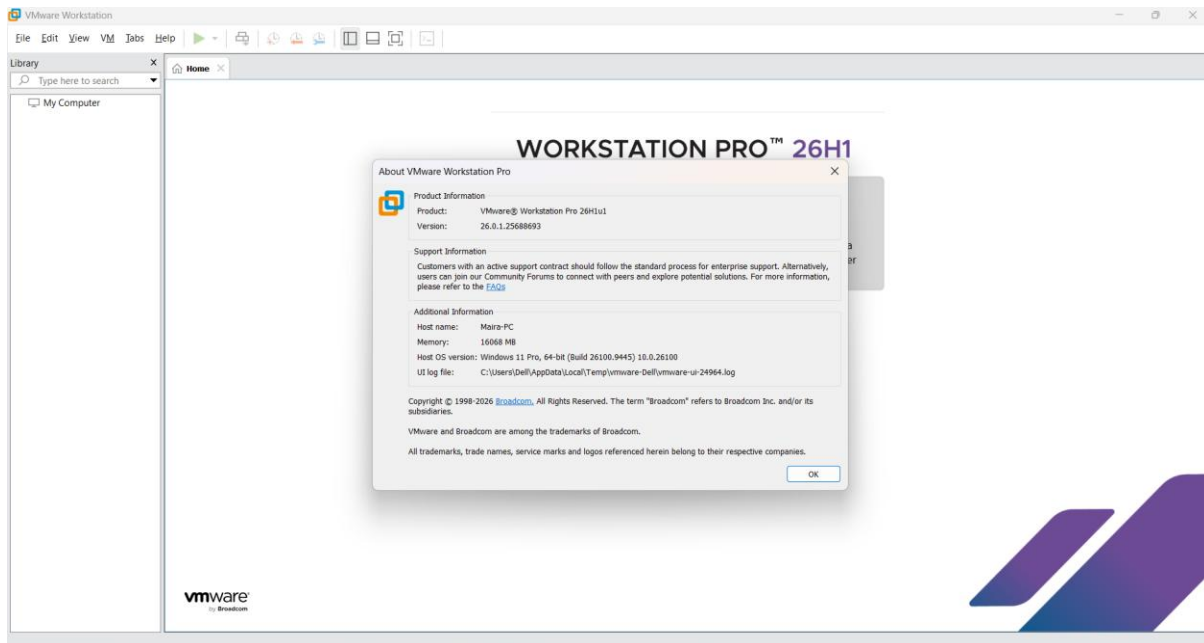
Cloud Computing (SE-665) Section B
Waiting for administrator approval

Pending approval

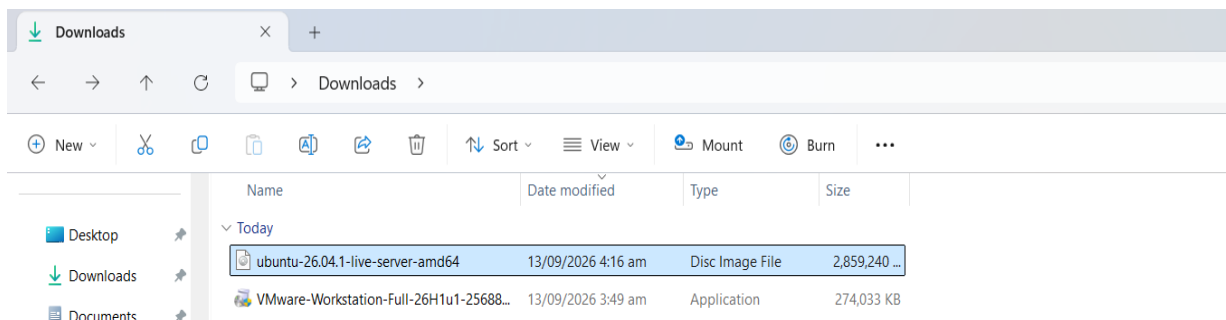
Task 3: Download VMware Workstation Pro and Ubuntu Server

Screenshots:

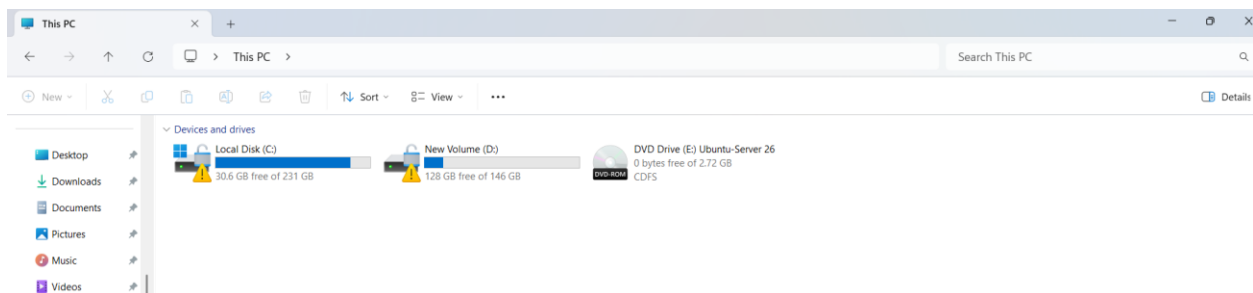
- vmware_workstation.png



- ubuntu_server_iso.png



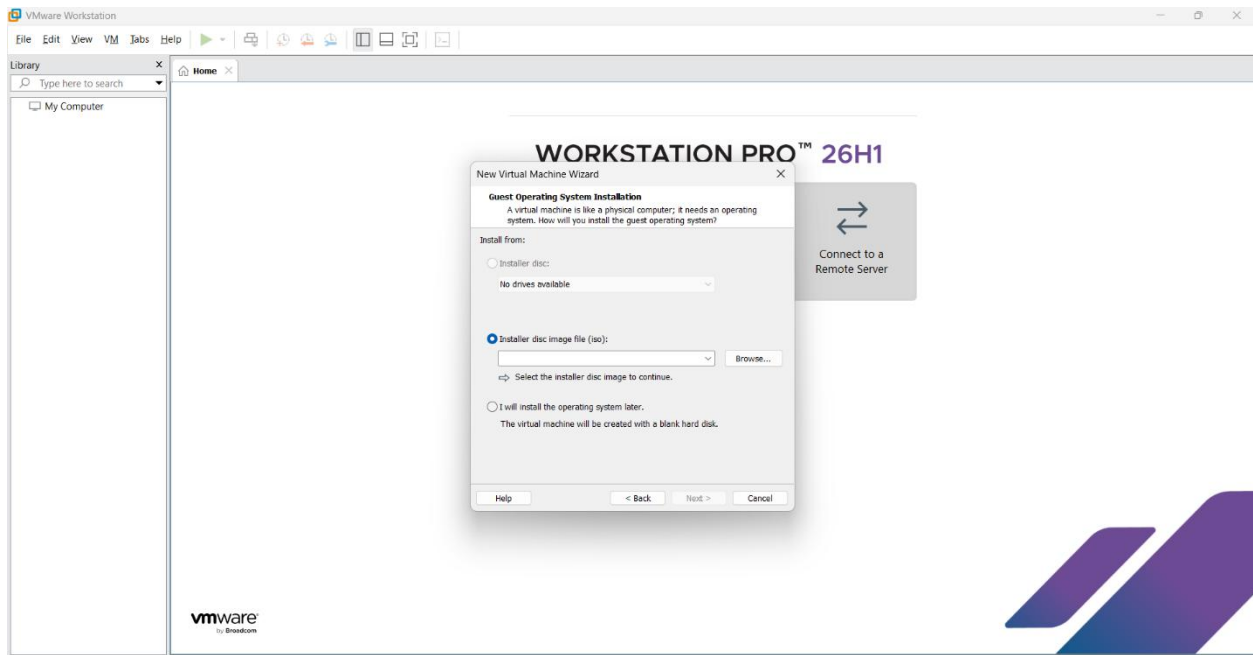
- available_storage.png



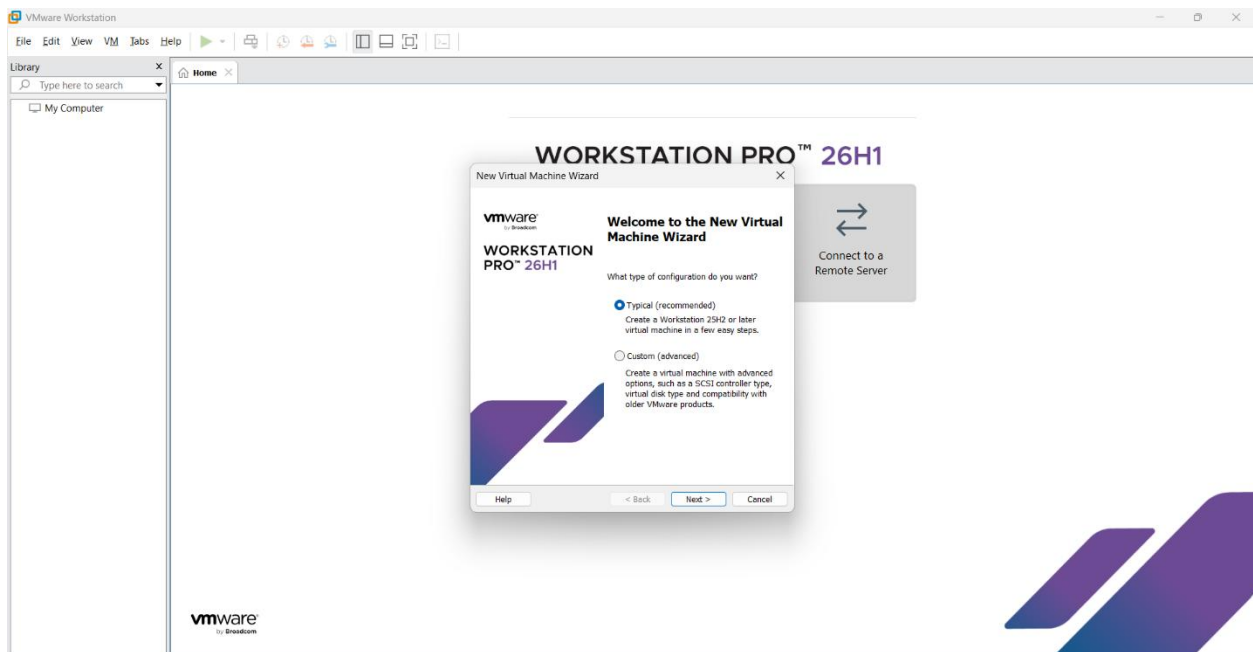
Task 4: Create the Ubuntu Server Virtual Machine

Screenshots:

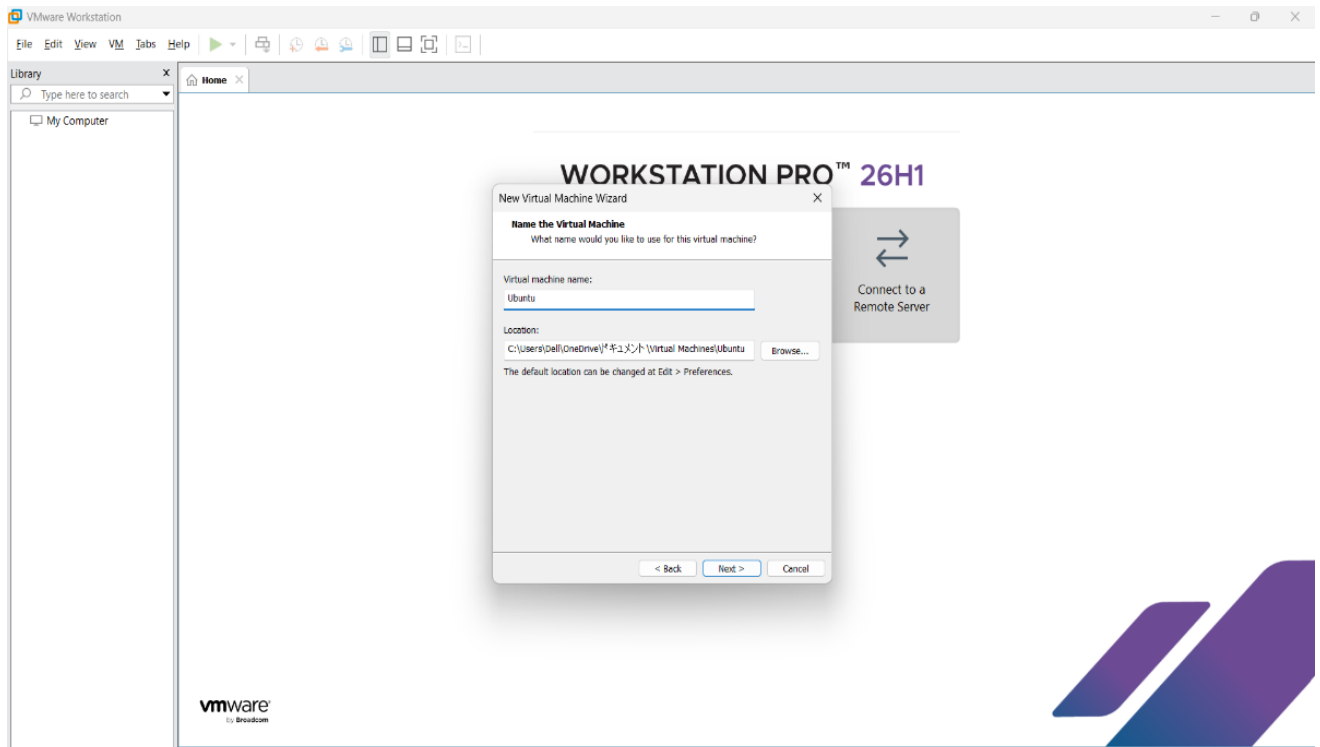
- vm_creation_wizard.png



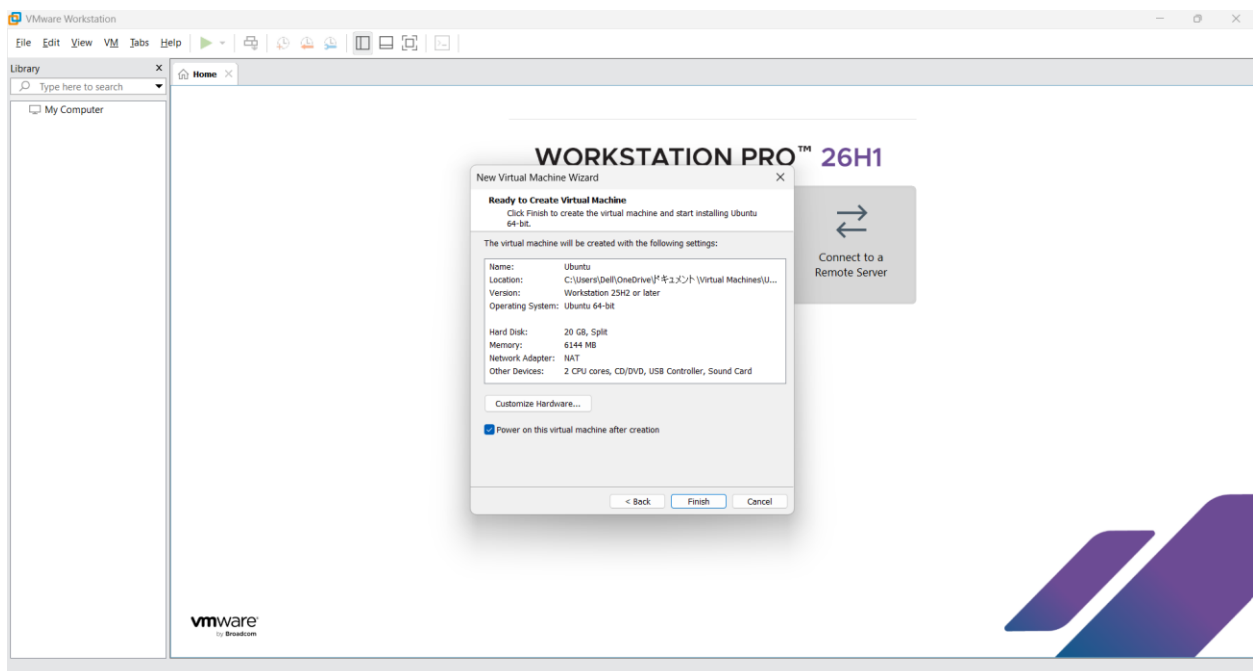
- vm_typical_configuration.png



- vm_ubuntu_iso_selected.png



- `vm_configuration_summary.png`

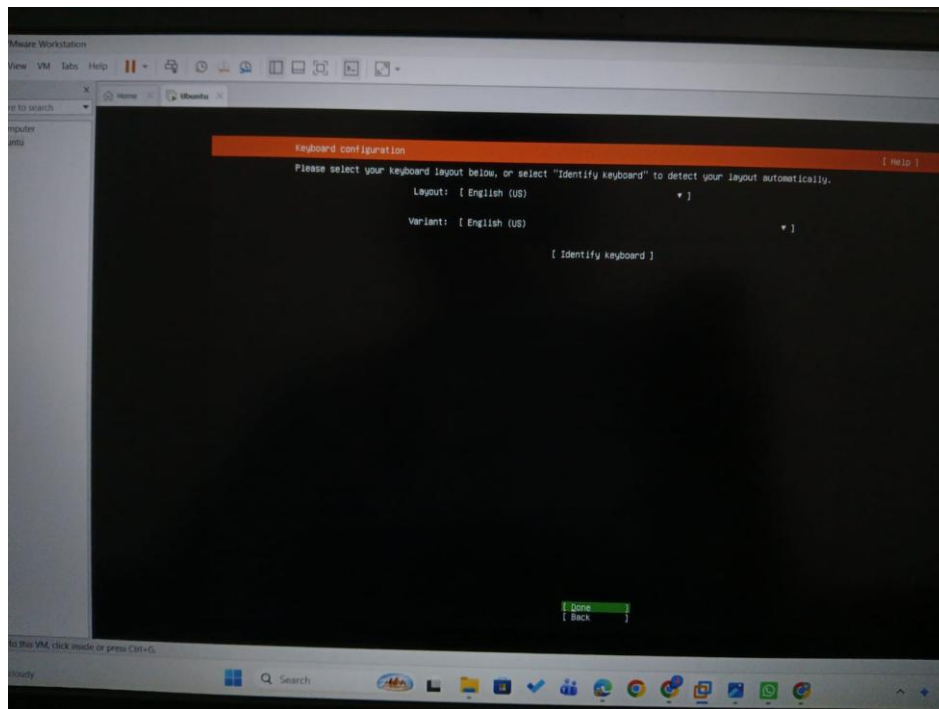


- `ubuntu_installer_boot.png`

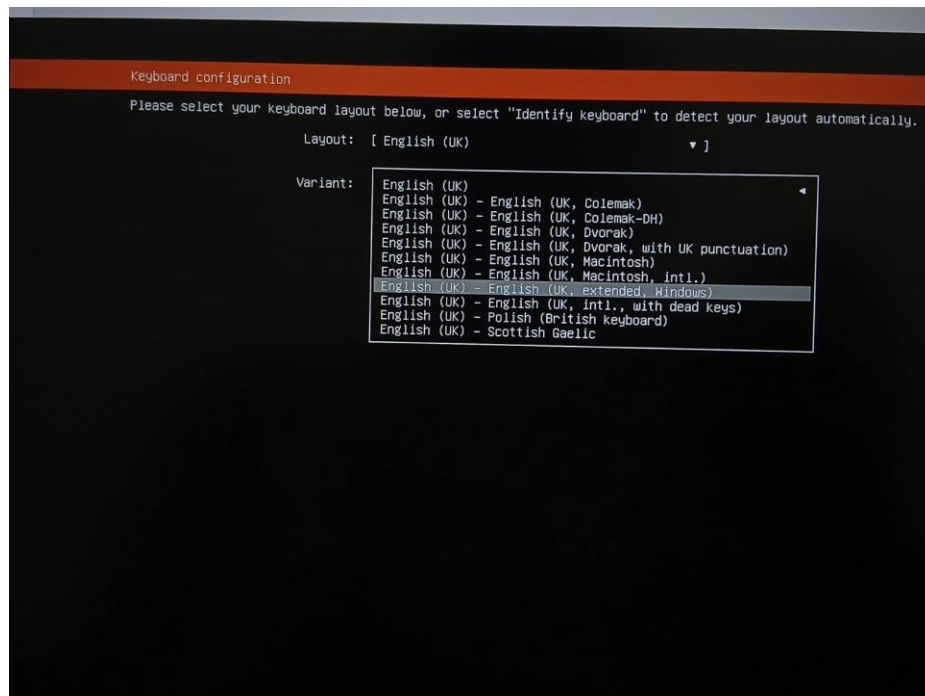
```
Starting polkit.service - Authorization Manager...
Starting rsyslog.service - System Logging Service...
[ OK ] Reached target getty-pre.target - Preparation for Logins.
Starting snapd.seeded.service - Wait until snapd is fully seeded...
Starting snapd.service - Snap Daemon...
Starting ssh.service - OpenBSD Secure Shell server...
Starting sysstat.service - Resets System Activity Logs...
Starting systemd-logind.service - User Login Management...
Starting udisks2.service - Disk Manager...
[ OK ] Finished dracut-shutdown.service - Restore /run/initramfs on shutdown.
[ OK ] Finished nvme-autoconnect.service - Connect NVMe-of subsystems automatically during boot
[ OK ] Reached target remote-fs-pre.target - Preparation for Remote File Systems.
[ OK ] Reached target remote-fs.target - Remote File Systems.
Starting apport.service - automatic crash report generation...
[ OK ] Started cron.service - Regular background program processing daemon.
Starting multipathd-queueing.service - Enable queuing for multipath maps...
Starting systemd-user-sessions.service - Permit User Sessions...
[ OK ] Finished sysstat.service - Resets System Activity Logs.
[ OK ] Finished multipathd-queueing.service - Enable queuing for multipath maps.
[ OK ] Started ssh.service - OpenBSD Secure Shell server.
[ OK ] Finished systemd-user-sessions.service - Permit User Sessions.
Starting plymouth-quit-wait.service - Hold until boot process finishes up...
Starting plymouth-quit.service - Terminate Plymouth Boot Screen...
[ OK ] Finished plymouth-quit-wait.service - Hold until boot process finishes up.
Starting setvtrgb.service - Set console scheme
Starting systemd-journal-catalog-update.service - Rebuild Journal Catalog
[ OK ] Started ugauth.service - Authenticating virtual machines hosts
[ OK ] Started open-vn-tools.service - Set up virtual machines host tools
[ OK ] Finished systemd-udev-trigger.service - Coldplug all udev devices
[ OK ] Started systemd-ask-password-console-requests.service - Local Encrypted Volumes
[ OK ] Reached target cryptsetup.target - Local Encrypted Volumes
[ OK ] Finished systemd-journal-catalog-update.service - Rebuild Journal Catalog
[ OK ] Finished netplan-configured.service - Netplan Backend Configuration
[ OK ] Finished snapd.apparmor.service - dofiles managed internal state
[ OK ] Finished console-setup.service - Set console font and keymap
[ OK ] Finished console-setup.service - Set console font and keymap
6.9.7-20230801 p11x4 smbus 0000:00:07.3: SMBus Host Controller not
```

Screenshots:

- ubuntu_keyboard_layout.png



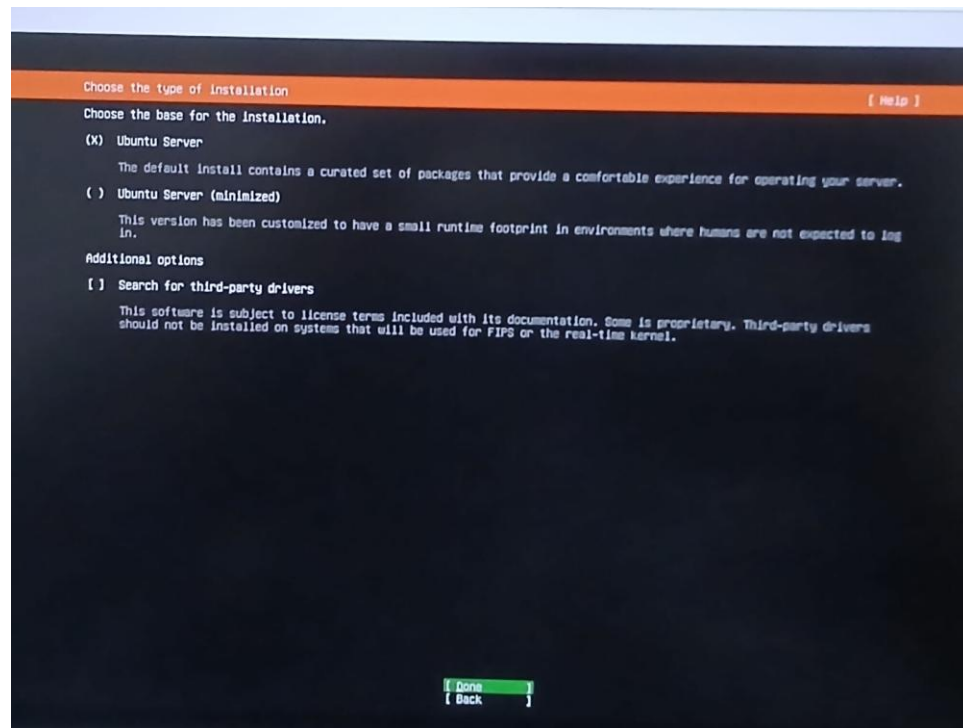
- ubuntu_keyboard_variant.png



Task 7: Choose the Ubuntu Server Installation Type

Screenshots:

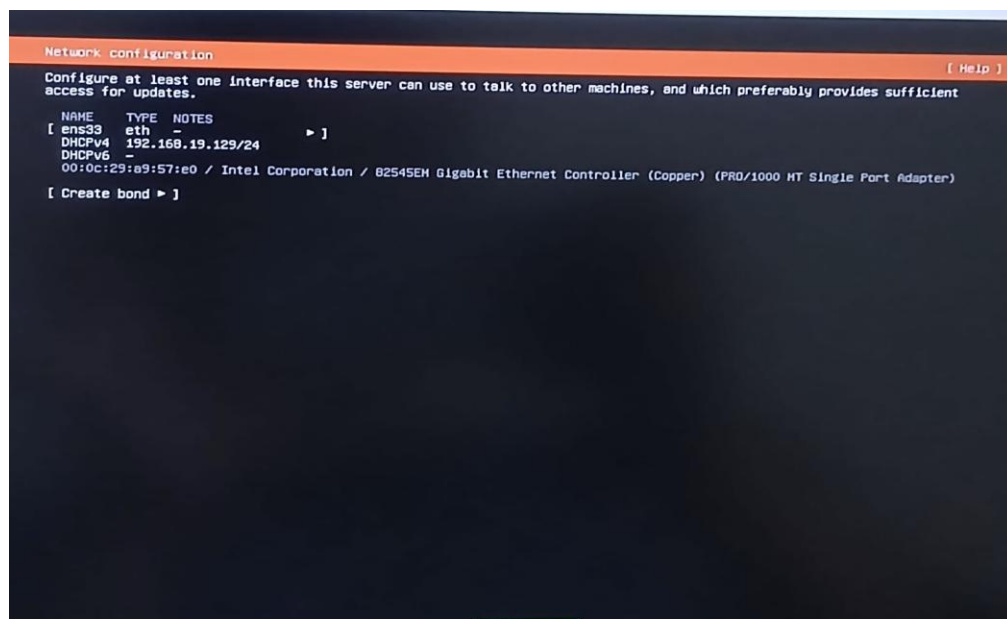
- ubuntu_installation_type.png



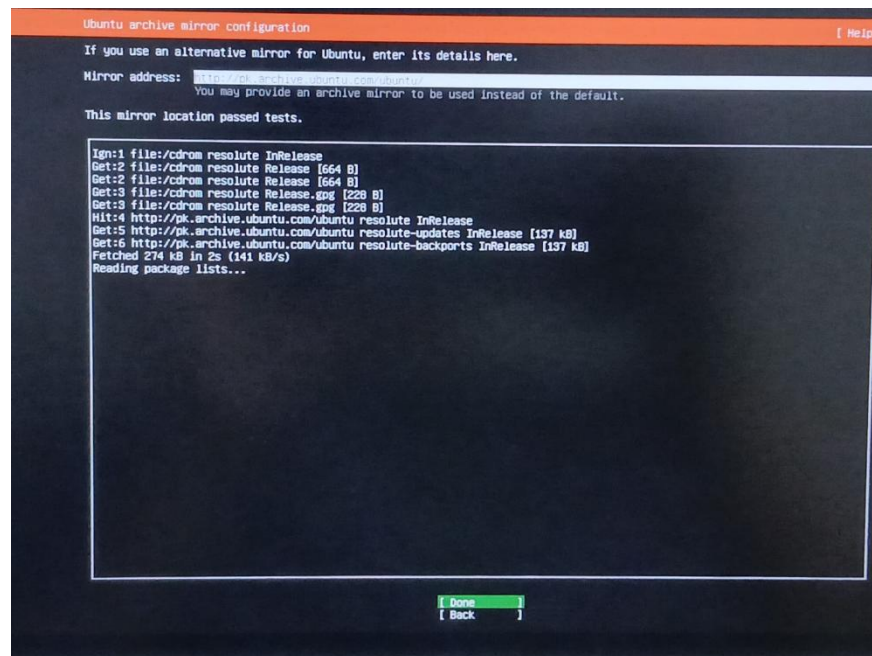
Task 8: Configure the Network

Screenshots:

- ubuntu_network_interface.png



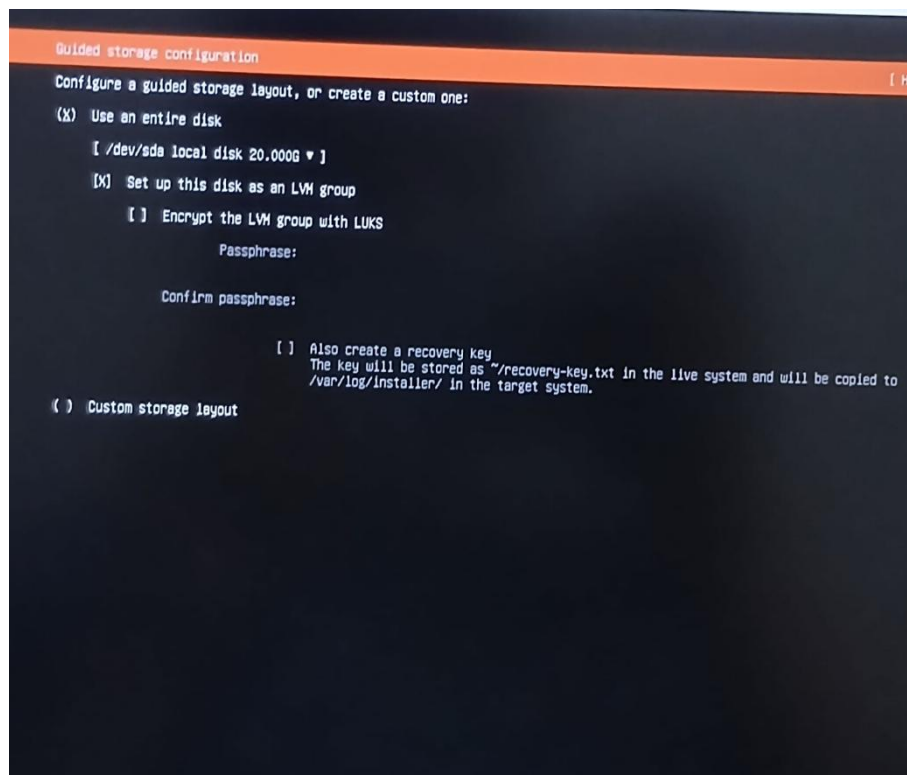
- ubuntu_installer_network.png



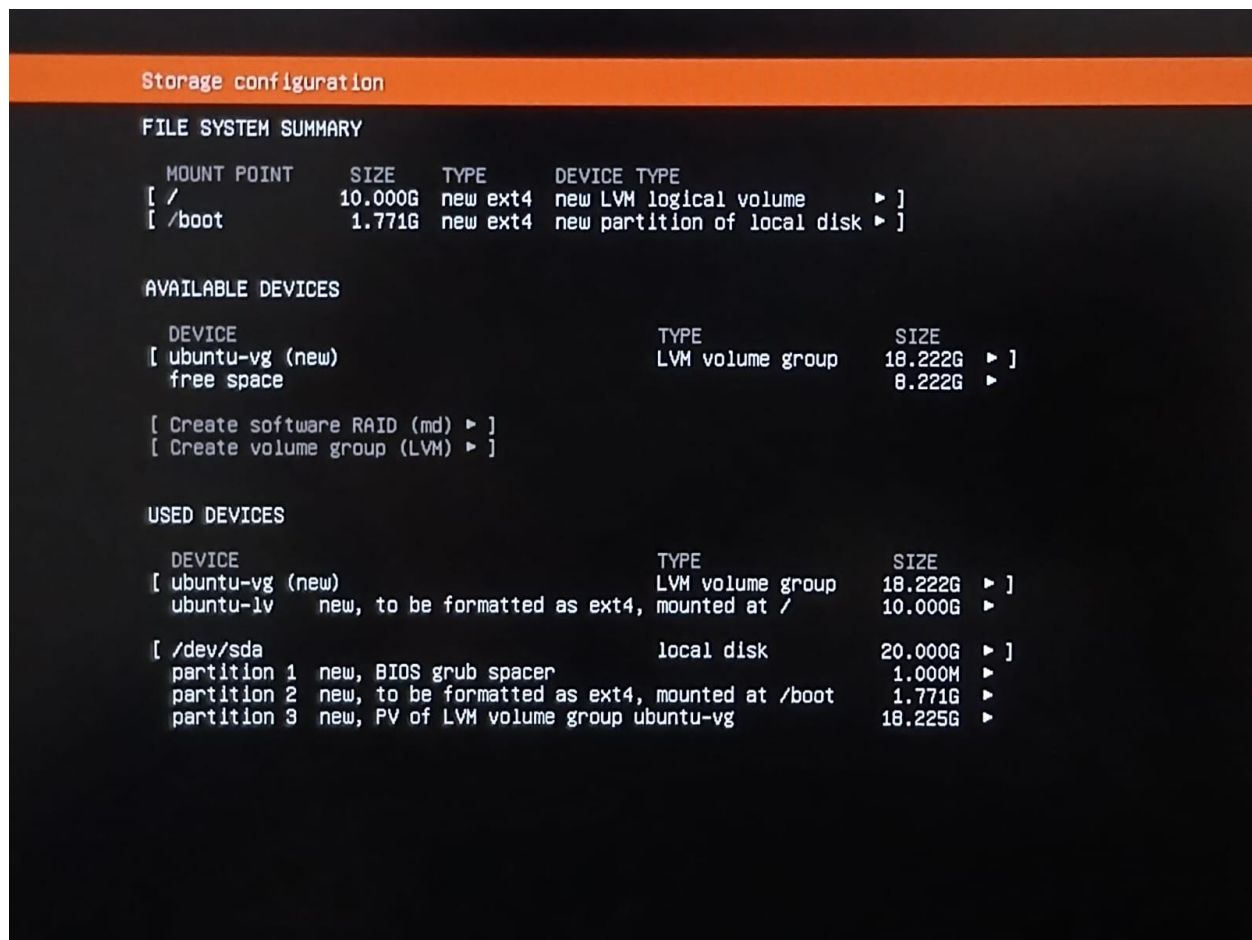
Task 9: Configure Storage

Screenshots:

- ubuntu_guided_storage.png



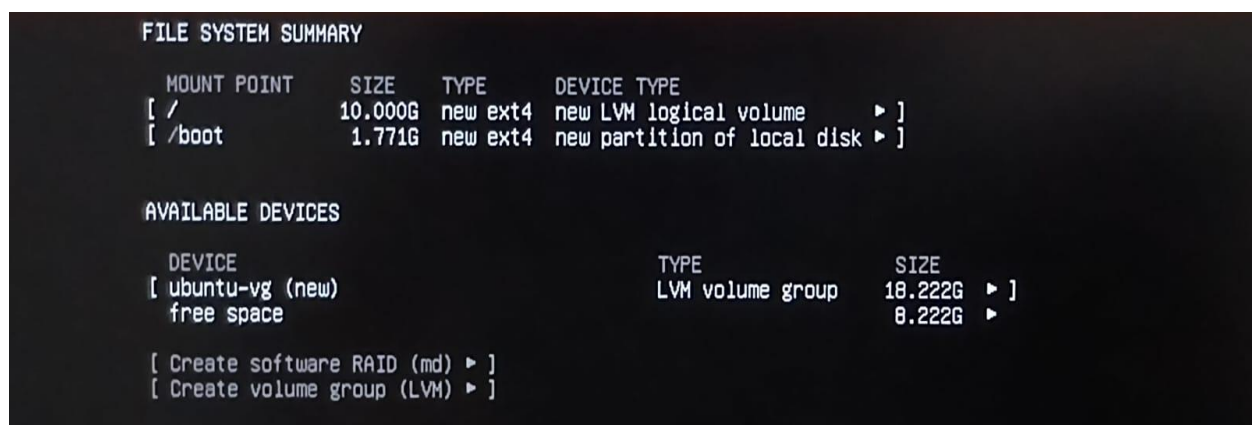
- ubuntu_storage_configuration.png



Task 10: Select the Installation Disk

Screenshots:

- ubuntu_selected_disk.png



- ubuntu_disk_details.png

```

FILE SYSTEM SUMMARY

  MOUNT POINT      SIZE    TYPE    DEVICE TYPE
[ /                10.000G new ext4 new LVM logical volume ▶ ]
[ /boot           1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

  DEVICE                                TYPE                SIZE
[ ubuntu-vg (new)                        LVM volume group    18.222G ▶ ]
  free space                             8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

  DEVICE                                TYPE                SIZE
[ ubuntu-vg (new)                        LVM volume group    18.222G ▶ ]
  ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ▶ ]

```

Task 11: Review the Partitions

Screenshots:

- ubuntu_partition_layout.png

```

Storage configuration

FILE SYSTEM SUMMARY

  MOUNT POINT      SIZE    TYPE    DEVICE TYPE
[ /                10.000G new ext4 new LVM logical volume ▶ ]
[ /boot           1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

  DEVICE                                TYPE                SIZE
[ ubuntu-vg (new)                        LVM volume group    18.222G ▶ ]
  free space                             8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

  DEVICE                                TYPE                SIZE
[ ubuntu-vg (new)                        LVM volume group    18.222G ▶ ]
  ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ▶ ]

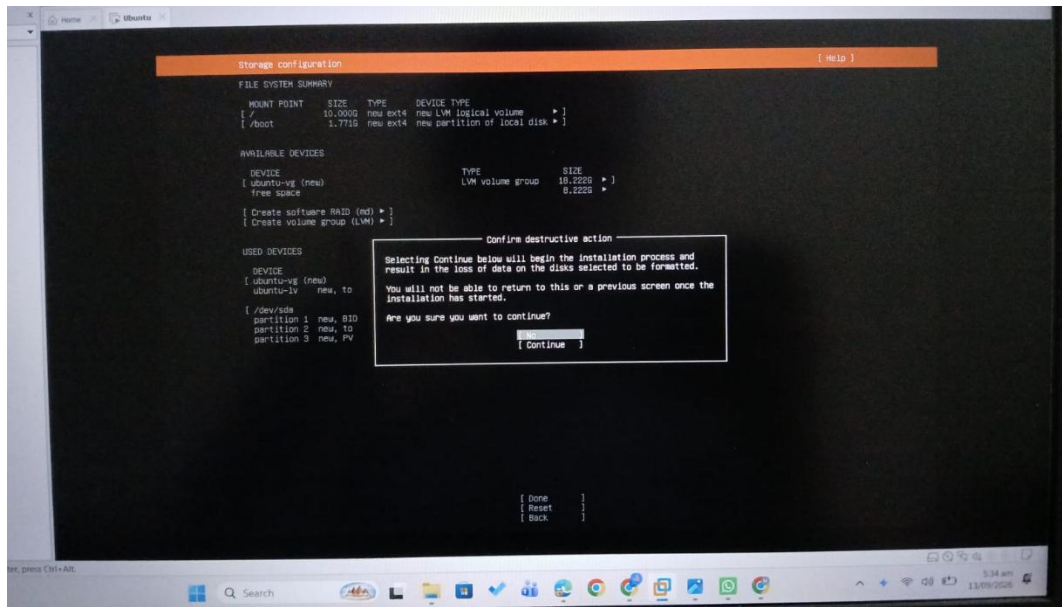
[ /dev/sda                                local disk          20.000G ▶ ]
  partition 1 new, BIOS grub spacer                    1.000M ▶ ]
  partition 2 new, to be formatted as ext4, mounted at /boot 1.771G ▶ ]
  partition 3 new, PV of LVM volume group ubuntu-vg        18.225G ▶ ]

```

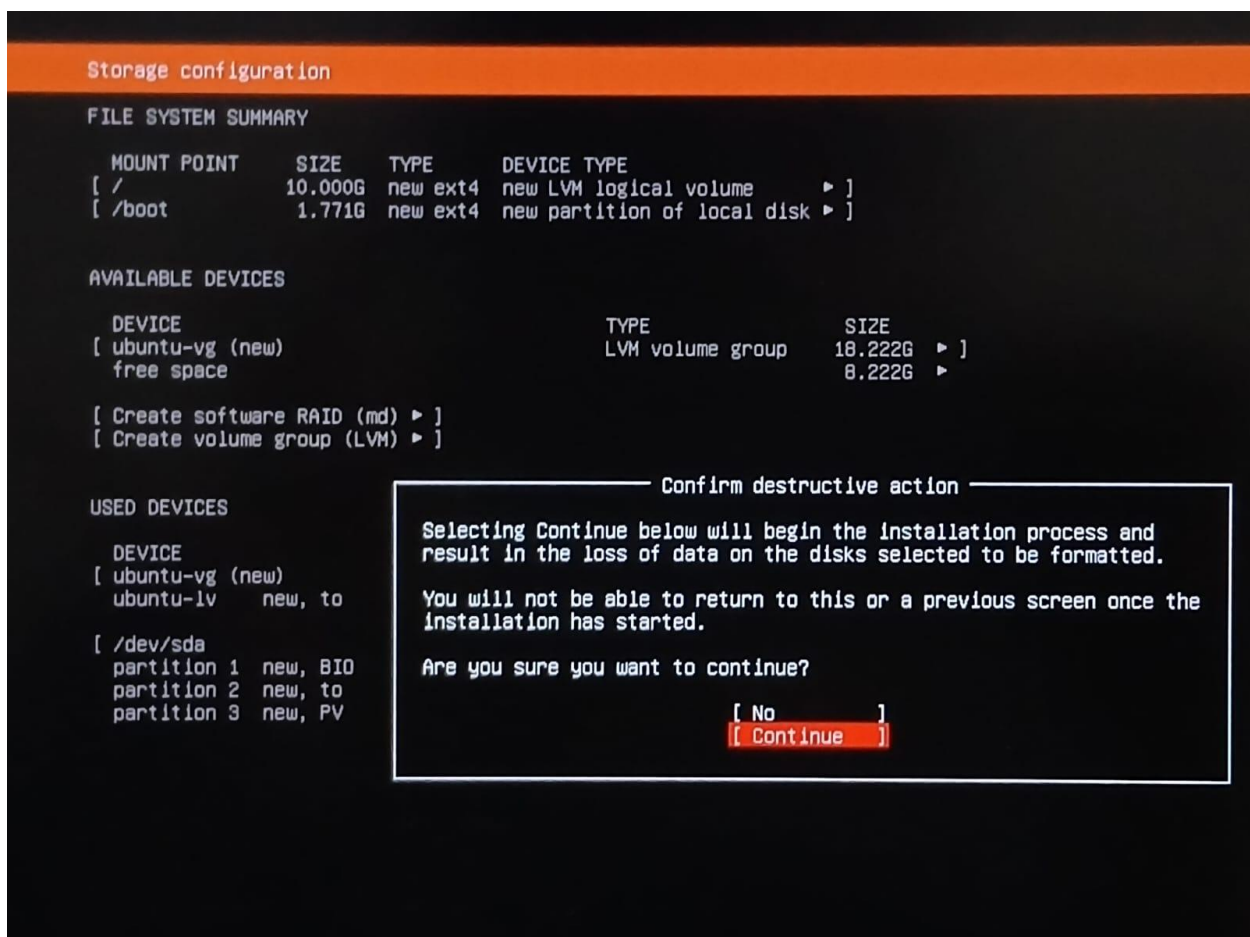
Task 12: Confirm the Storage Changes

Screenshots:

- ubuntu_storage_warning.png



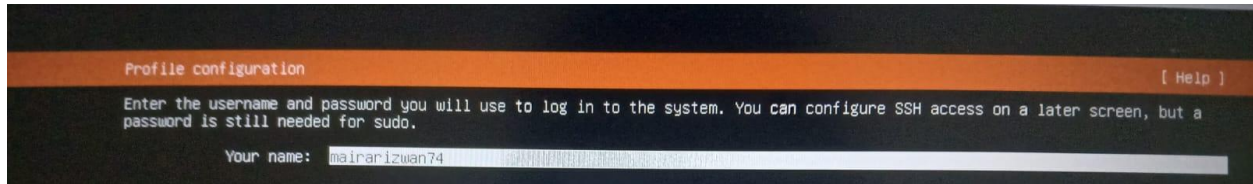
- ubuntu_storage_confirmed.png



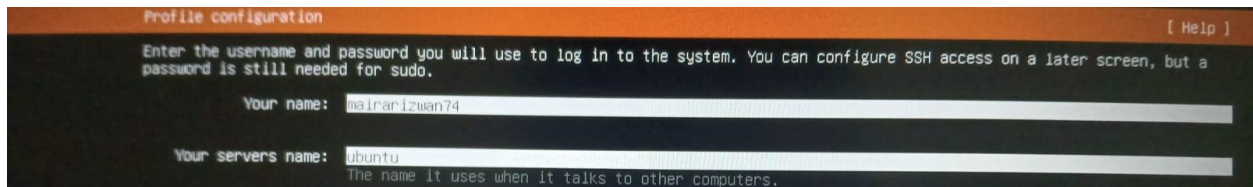
Task 13: Configure the Ubuntu Profile

Screenshots:

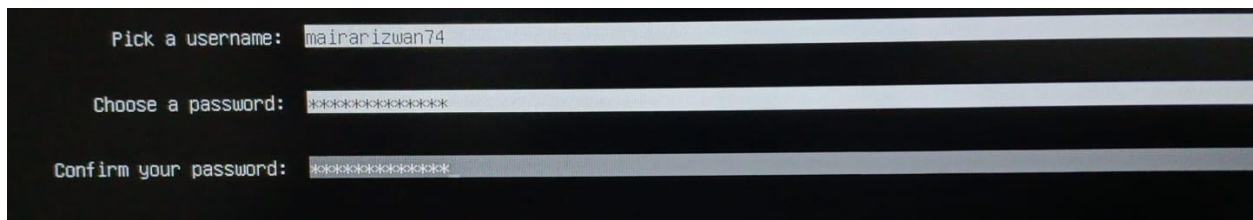
- ubuntu_actual_name.png



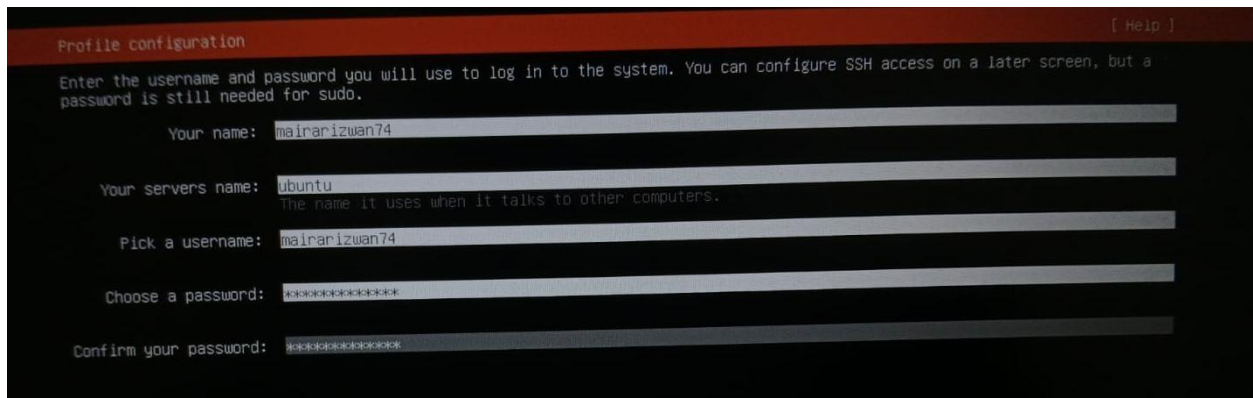
- ubuntu_server_name.png



- ubuntu_username.png



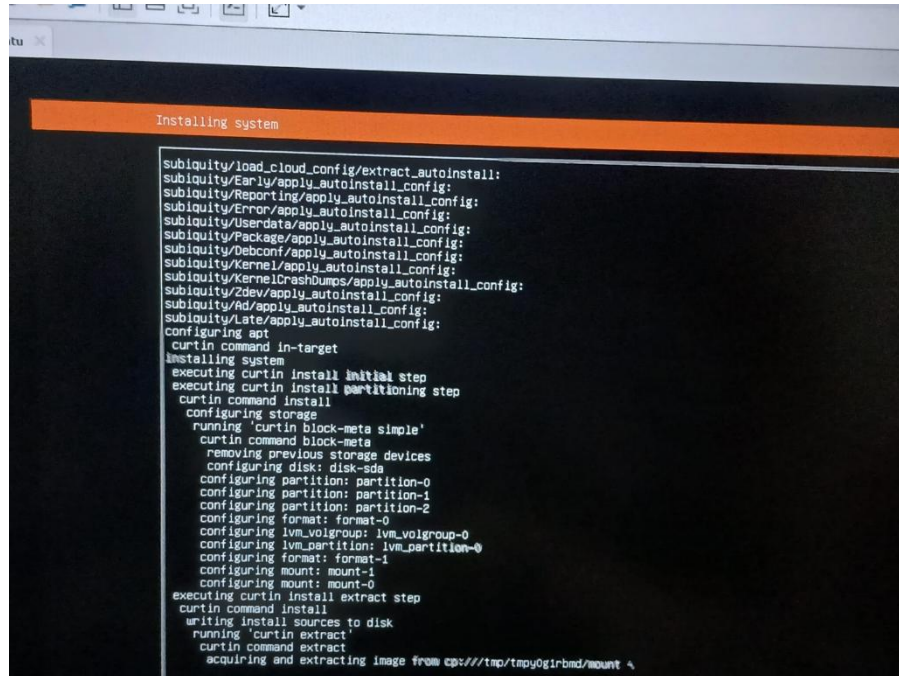
- ubuntu_profile_setup.png



Task 14: Complete the Software Installation

Screenshots:

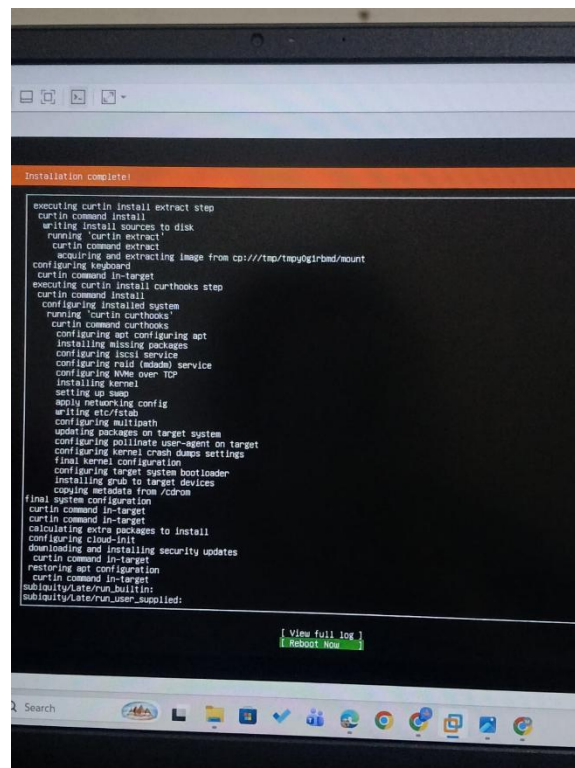
- ubuntu_installation_progress.png



Task 15: Reboot and Verify the Ubuntu Identity

Screenshots:

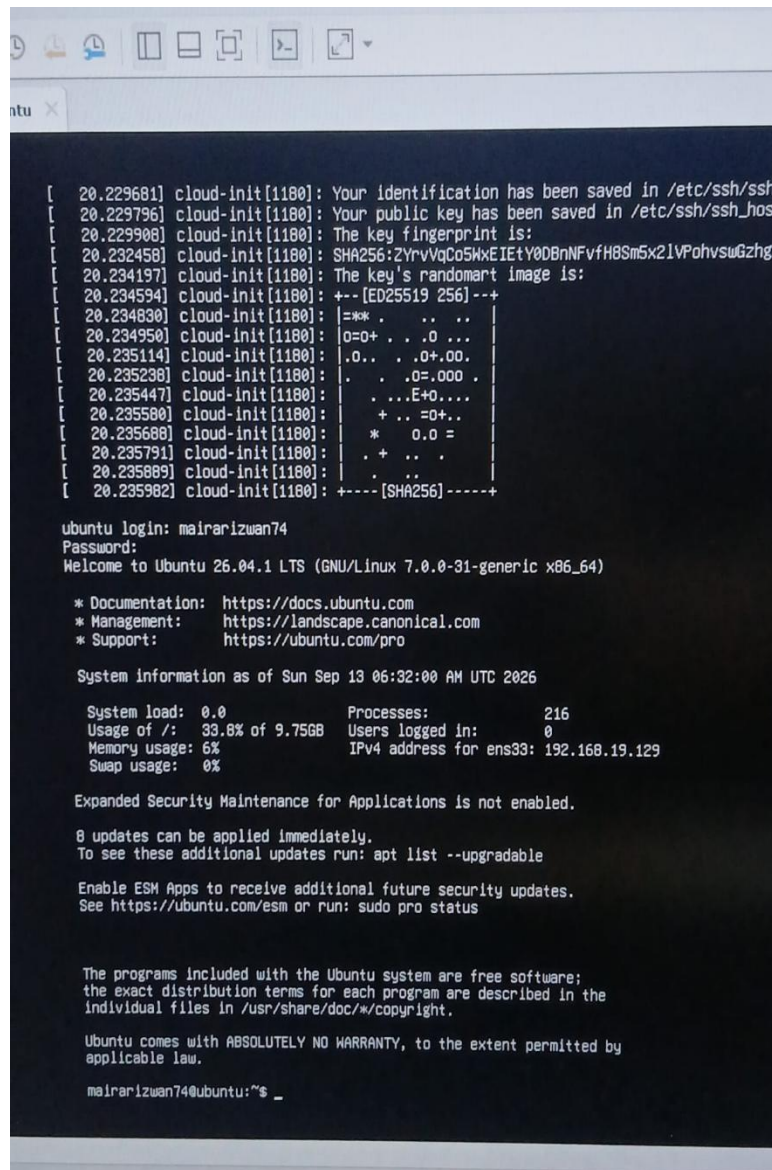
- ubuntu_installation_complete.png



- ubuntu_login_screen.png



- ubuntu_terminal_login.png



- ubuntu_identity_verified.png



Task 16: Find the Ubuntu Server IP Address

Screenshots:

- ubuntu_ip_addr_command.png

```
mairarizwan74@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:a9:57:e0 brd ff:ff:ff:ff:ff:ff
```

- ubuntu_ip_address.png

```
mairarizwan74@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:a9:57:e0 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29a957e0
    inet 192.168.19.129/24 metric 100 brd 192.168.19.255 scope global dynamic ens33
        valid_lft 1634sec preferred_lft 1634sec
    inet6 fe80::20c:29ff:fea9:57e0/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
mairarizwan74@ubuntu:~$ ssh -l mairarizwan74 192.168.19.129
The authenticity of host '192.168.19.129 (192.168.19.129)' can't be established.
ED25519 key fingerprint is: SHA256:ZYrvVqCo5WxEIEtY0DBnNFvfH8Sm5x2lVPohvswGzhg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes_
```

Task 17: Connect to Ubuntu Server from Windows

Screenshots:

- windows_ssh_command.png

```
C:\Users\Dell>ssh mairairizwan74@192.168.19.129
```

- windows_ssh_fingerprint.png

```
C:\Users\Dell>ssh mairairizwan74@192.168.19.129
The authenticity of host '192.168.19.129 (192.168.19.129)' can't be established.
ED25519 key fingerprint is: SHA256:ZYrvVqCo5WxEIEtY0DBnNFvfH8Sm5x2lVPohvswGzhg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

- windows_ssh_login.png

```
mairarizwan74@ubuntu: ~  
Microsoft Windows [Version 10.0.26100.9445]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dell>ssh mairarizwan74@192.168.19.129  
mairarizwan74@192.168.19.129's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Sun Sep 13 08:38:16 AM UTC 2026  
  
System load:  0.01      Processes:            220  
Usage of /:   34.2% of 9.75GB  Users logged in:    0  
Memory usage: 7%      IPv4 address for ens33: 192.168.19.129  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
7 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Last login: Sun Sep 13 06:55:58 2026 from 192.168.19.129  
mairarizwan74@ubuntu:~$
```

- windows_ssh_identity.png

```
mairarizwan74@ubuntu:~$
```